YEAR 8 · SCIENCE DETECTIVES 科学侦探 · WEEK 1 OF 10 · 40 MINUTES

Think like a detective 像侦探一样思考

Detectives and scientists separate what they see (observations) from what they think it means (inferences), and trust evidence over opinion.

- I can sort statements into observations and inferences.
- I can write an observation and an inference about the scene, and say what evidence checks it.

forensic science 法庭科学 · detective 侦探 · evidence 证据 · crime scene 案发现场 · suspect 嫌疑人 · observation 观察 · inference 推断 · opinion 观点 · alibi 不在场证明

THE LESSON

Do Now: skills check	15	Seat fixed groups on arrival; read instructions once; 15 minutes alone, 5-minute warning
Do Now: interest survey	3	Collect skills checks; 6-question smiley survey; show Today slide (goals, key words)
Hook	4	Case alert: briefing and note on Slides 5 and 6; Song Ci story in one minute; open case files
Learn	4	Observation vs inference (glass); eye = observe, tap head = infer; O or I for 3 statements
Do	10	Say, show, check (Slide 9); scene map; Core (O or I, 8 statements), Stretch, Challenge
Think and Reflect	4	Meet the four suspects; thumbs vote (opinion); exit ticket: one O, one I; worksheet into file

Hands-on: No practical: paper scene map task, with an optional look-only scene table at the front.

Collect or assess: Skills Check baseline (out of 20) and survey into tracker; filed, not returned before Week 10. Exit ticket: O vs I.

SAFETY

- No practical hazards beyond normal classroom rules.
- Optional scene table: look only. No touching, tasting or smelling the evidence.
- Hot water for the optional scene liquid is teacher-only.

WATCH OUT

- Start the skills check as students sit; it takes the full 15 minutes. Never return papers before Week 10.
- Guesses get called observations ('The powder is sugar'). Ask: 'Can you see that, or are you guessing?'

Language: Frames: 'I observe ___ at square ___.' 'I infer ___ because ___.' Chinese pair talk welcome; Band 3 buddy for Band 1.

Next week: Week 2: fingerprints. Every case file needs its suspect cards. Keep pens A to D, strips and note for Week 3.

PREP FOR ONE CLASS

before the lesson about 220 min (155 could be done by someone else); after about 105 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Order the kit; plain soap and baking soda only if using the scene table 15 min · technician
- [ ] Book printing for Weeks 1 to 5; freeze Skills Check master until Week 10 15 min · teacher
- [ ] Print, fold and staple dictionaries x28 and logbooks x24 30 min · reprographics

THE WEEK BEFORE

- [ ] Print language pack (4 posters, cards) and worksheets x24 (48 sheets) 20 min · reprographics
- [ ] Laminate and cut role and language cards; cut frame cards; label class box 55 min · technician
- [ ] Pen pre-test (pens A to D, record, keep strips); write the note with pen B 25 min · teacher
- [ ] Print teacher copies; get allergy list (needed for Week 4 powders) 10 min · teacher
- [ ] Plan 6 groups of 4 (Band 1 beside a Band 3 buddy); make name cards 15 min · teacher
- [ ] Print scene table sign and 24 cm footprint at 100%; check with a ruler optional 5 min · technician

THE DAY BEFORE

- [ ] Collate 24 case files; order checks, cut surveys; posters up; set tables 35 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Make the scene liquid and set up the look-only scene table optional 15 min · teacher

AFTER THE LESSON

- [ ] Clear the scene table; keep the note and pens A to D for Week 3 optional 5 min · technician
- [ ] Mark the skills checks within the week (no half marks; accept Chinese) 70 min · teacher
- [ ] Enter tracker, file papers; read exit tickets; check case files in box 35 min · teacher

PRINT

- Skills Check x24 (3 sheets each, double-sided; length to confirm)
- Worksheet x24 (2 sheets each); interest survey x24 (half A4, cut)
- Briefing (double-sided), scene map, suspect cards x24 (1 sheet each)
- Case file cover x24, evidence log x24 (A4 card if possible)
- Dictionary x28 (A5 booklet, 7 sheets each, 90 gsm); logbook x24 (A5)
- Wall: A3 scene map, 3 A3 language posters, A4 frame poster
- Frame cards x6 sets (4 sheets each); role cards x6 sets (1 sheet each)
- Classroom language cards x30 (A5, double-sided, laminated)
- Teacher copies: kit card, mark scheme, pre-test record
- Scene table sign (A5), true-size footprint (A4 at 100%) (optional)

GET OR MAKE (NEW THIS LESSON)

- Black washable felt-tips x8 (4 brands, 2 of each)
- Coffee filters (4 strips, 2 x 10 cm), cups x4, water
- Masking tape x1 roll
- White A4 card x49 (covers, evidence logs, note)
- A4 folders x24 (collated into case files)
- Pencils x24
- Laminating pouches: A4 x6, A5 x30
- Class box x1 (a section per group)
- Class assessment folder x1 (may already exist)
- Name cards (number not stated; 24 assumed)
- Make: pen pre-test strips x4, display note
- Scissors or guillotine, laminator, ruler and pencil
- Scene table: glass, dark tray, saucer (optional)
- Bar soap, baking soda, grater, kettle, jug (optional)

Have ready from earlier weeks: Nothing.